CROSS-ASSET VOLATILITY CONVERGENCE

A Regime-Conditional Macro Playbook (2008 - 2026)

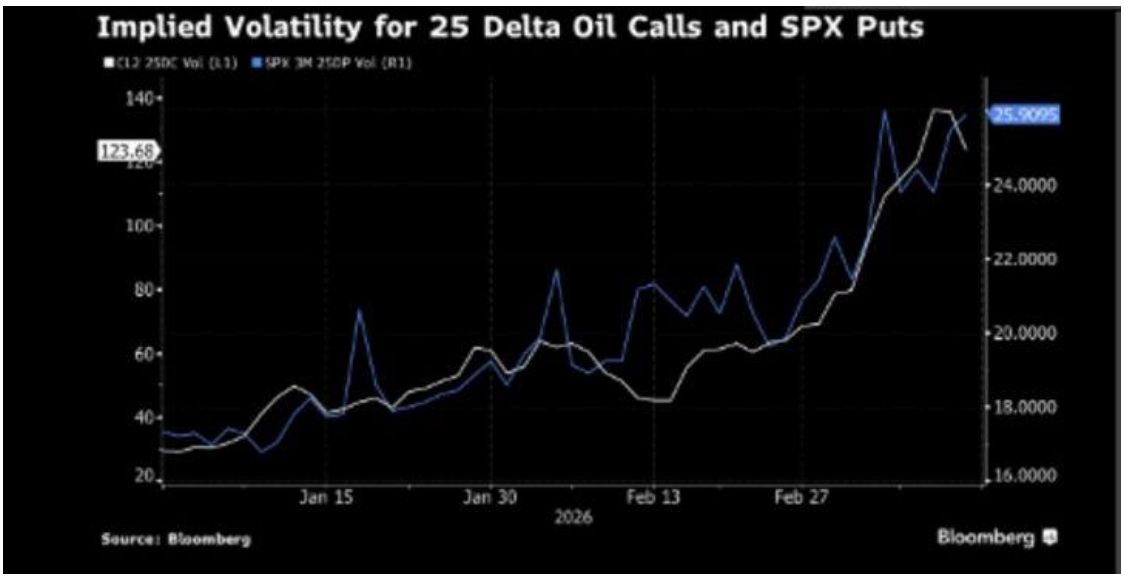
Israel Olujobi | April 2026

GitHub: github.com/Israel-Olujobi/cross_asset_vol_convergence

Executive Summary

This research builds a regime-conditional cross-asset volatility convergence framework which was motivated by a Bloomberg chart showing crude oil 25-delta call volatility and SPX put volatility converging during the current Iran conflict in early 2026.

The central question: when oil and equity volatility converge above a critical threshold, does the nature of the leading stress determine the subsequent cross-asset response?



Img.1: Implied Volatility for 25 Delta Oil Calls and SPX Puts - Bloomberg.

The answer, validated across 18 years of data and 14 convergence episodes, is yes. The framework identifies two distinct stress regimes, oil-led and equity-led, each producing structurally different cross-asset responses. Getting the regime wrong means executing the opposite trade. The live Iran war test confirmed signal activation on March 6th 2026, achieving 77% directional accuracy across non-structurally distorted assets.

1. Motivation & Research Question

As the Iran crisis was properly taking off in early March 2026, the Bloomberg chart observed revealed a striking convergence between crude oil implied volatility (OVX) and SPX volatility (VIX). This convergence, occurring above historical stress thresholds, raised a question that sits at the intersection of commodity markets, equity volatility and macro regime analysis:

"When oil volatility and equity volatility converge above a critical threshold, does the nature of the leading stress, oil-led vs equity-led, determine the subsequent cross-asset response?"

This question motivated the construction of a complete analytical pipeline: from signal detection and regime classification through event study analysis, Central Bank sentiment integration, XGBoost modelling and live signal assessment.

2. Data & Methodology

2.1 Dataset

23 assets across equities, fixed income, commodities, FX and credit, sourced via yfinance. Daily data from May 2007 to March 2026 (4,916 rows), as OVX was launched in yfinance in May 2007. Assets include SPX, FTSE, EuroStoxx, Nikkei, Gold, TLT, WTI, Brent, OVX, VIX, HYG, DXY, US10Y and 10 currency pairs.

2.2 Convergence Signal

Two signal types were constructed from 252-day rolling Z-scores:

- Type 1: 60-day VIX/OVX rolling correlation > 0.6 AND both Z-scores > 1.0σ
- Type 2: Both Z-scores > 2.0σ simultaneously

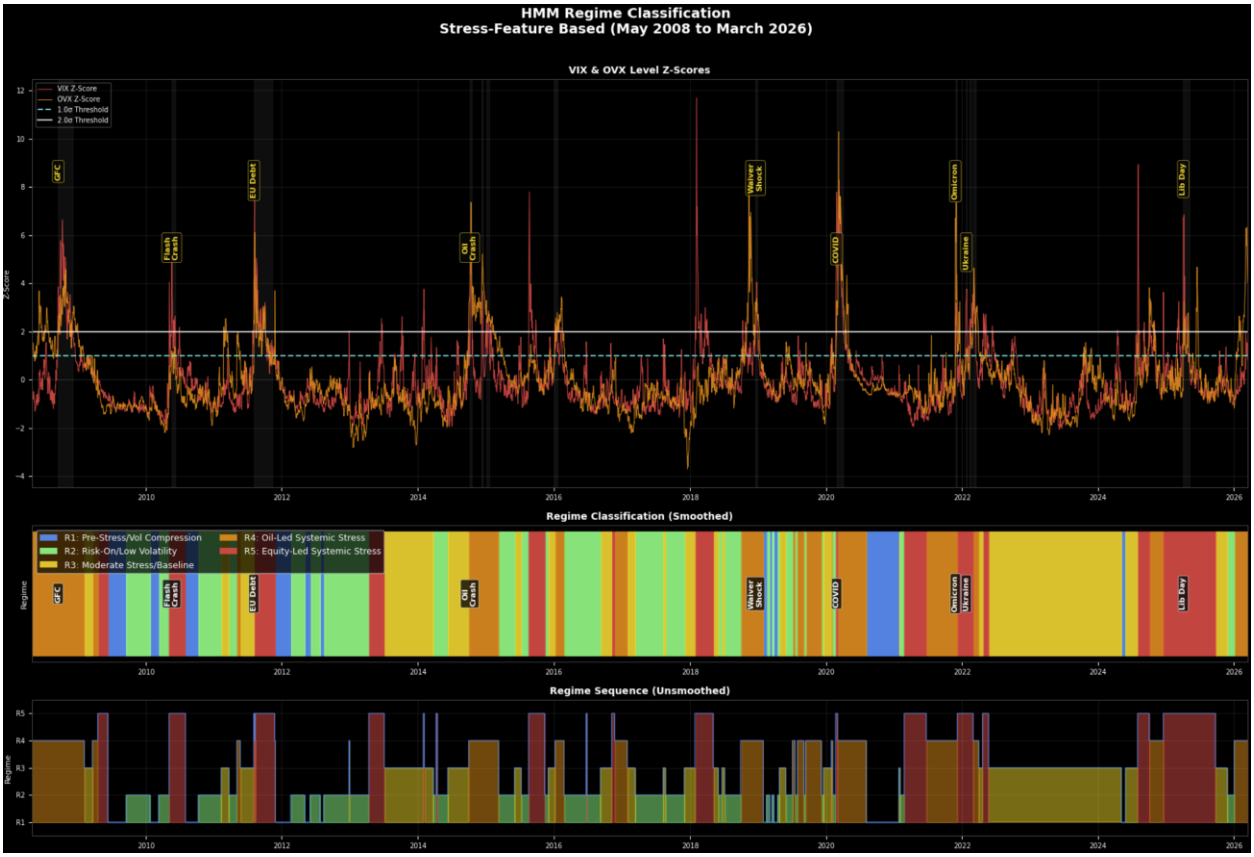
14 convergence episodes were identified between 2008 - 2026 using a 5-day gap tolerance (trading days allowed within same episode that don't meet criteria) and 5-day minimum duration filter (filtering out noise).

Img.2: Cross-Asset Vol Convergence Signal.

2.3 Regime Classification

A 5-state Hidden Markov Model (HMM) was trained on 8 stress features. Two systemic stress regimes were identified:

- Regime 4: Oil-Led Systemic Stress: OVX leading VIX, 9 convergence episodes
- Regime 5: Equity-Led Systemic Stress: VIX leading OVX, 5 convergence episodes



Img.3: VIX & OVX convergence and HMM Regime Classification.

2.4 Central Bank Sentiment Layer

154 FOMC statements (2008 - 2026) were scored using ProsusAI/FinBERT. Rate decisions and basis points were sourced from FRED API. The three central bank signal features integrated included:

- cb_sentiment (FinBERT score -1 to +1)
- cb_decision (rate direction: +1 hike, -1 cut, 0 hold)
- cb_rate_bp (magnitude in basis points), each forward-filled daily across 4,662 trading days.

The decision distribution identified 3,600 hold trading days (77%), 656 hike trading days (14%) and 406 cut trading days (9%) in total between the time period.

2.5 Event Study

Forward returns were calculated at 30, 60 and 90-day horizons for 17 assets across all 14 convergence episodes resulting in 714 total observations. Hit rates were calculated as the percentage of episodes where the asset achieved a positive return.

2.6 Model Comparison

XGBoost model with SHAP feature importance was compared with Logistic Regression's performance on the dataset. They used a three-class target variable: Class 0 (No Signal, 4434 days, 95.1%), Class 1 (Oil-Led Signal, 130 days, 2.8%), Class 2 (Equity-Led Signal, 98 days, 2.1%). Walk-forward validation was applied across the three expanding training windows (2008 - 2018, 2008 - 2020, 2008 - 2022), to prevent lookahead bias.

3. Key Findings

3.1 Regime-Conditional Cross-Asset Responses

The event study reveals fundamentally different asset responses across the two stress regimes:

Asset	Oil-Led 30d	Hit Rate	Oil-Led Signal	Equity-Led 30d	Hit Rate	Equity-Led Signal
Gold	+2.13%	78%	LONG*	+5.80%	100%	LONG*
WTI	-10.52%	33%	SHORT*	+15.39%	100%	LONG*
Brent	-8.73%	44%	SHORT*	+15.71%	80%	LONG*
DXY	+2.40%	67%	LONG*	+1.00%	60%	LONG*
TLT	+0.26%	56%	LONG	-1.10%	40%	SHORT*
USDBRL	+3.11%	56%	LONG	-1.73%	40%	SHORT*
SPX	-0.94%	67%	SHORT	+2.43%	60%	LONG*
NIKKEI	-4.67%	33%	SHORT*	-2.76%	40%	SHORT*
EUROSTOXX	-0.93%	67%	SHORT	-4.33%	20%	SHORT*
HYG	-1.42%	44%	SHORT*	0.79%	60%	LONG*
USDCHF	-1.88%	33%	SHORT*	0.86%	60%	LONG*
AUDUSD	-2.66%	22%	SHORT*	1.75%	80%	LONG*

The regime-conditional playbook turns the event study findings into directional signals across three time horizons. LONG* and SHORT* denote high-conviction signals where mean return exceeds 0.5% and hit rate meets the conviction threshold (≥60% for long, ≤45% for short). LONG and SHORT denote directional signals with lower conviction.

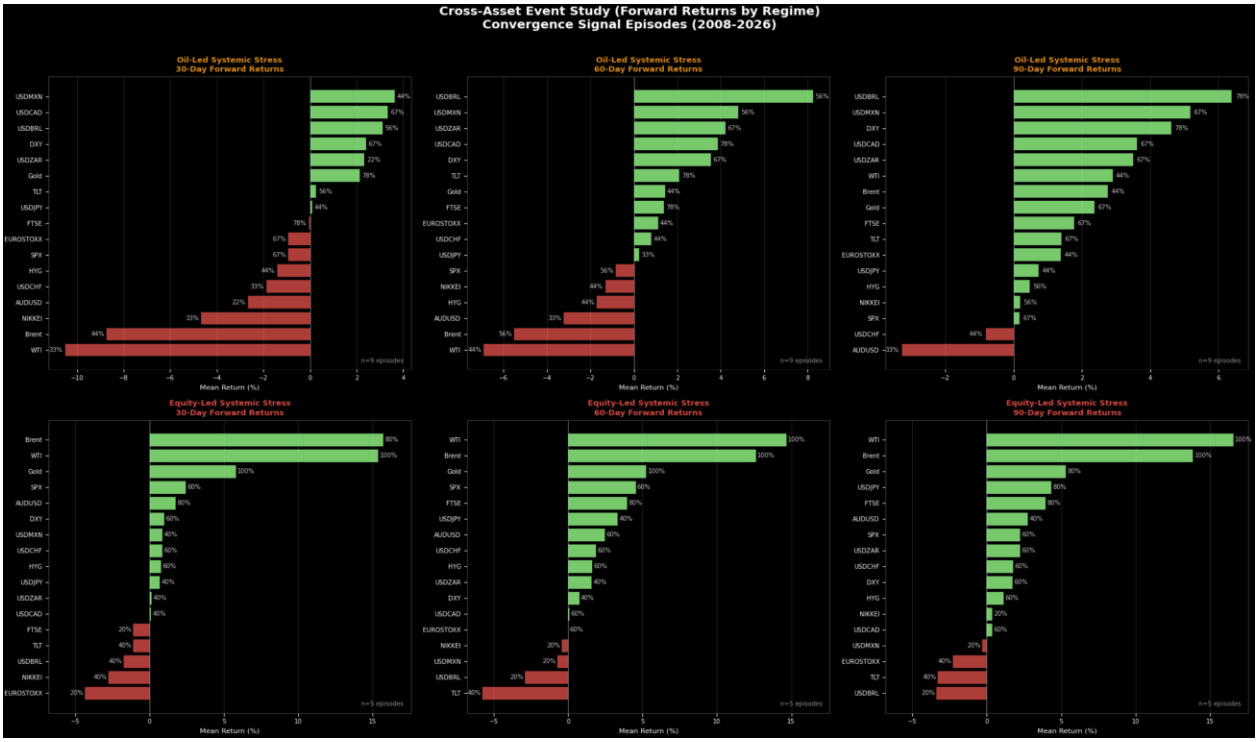
The regime distinction is not cosmetic, WTI is a SHORT* signal in oil-led stress (-10.52%) and a LONG* with 100% hit rate in equity-led stress (+15.39%). TLT strengthens in oil-led episodes and

sells off in equity-led ones, both episodes have the 60-day mark as peak in both directions. **Getting the regime wrong means executing the opposite trade.**

Gold & DXY are the most consistent cross-regime signal, LONG* in both oil-led (Gold - 78%, DXY - 67% hit rate) and equity-led stress (Gold 100% hit rate at 30 days, DXY - 60%). EM currencies weaken persistently in oil-led stress, with DXY strengthening from +2.40% at 30 days to +4.61% at 90 days.

In oil-led stress environments, the WTI crashes at 30 days (-10.52%) but fully recovers by 90 days (+2.90%), reflecting the mean-reverting nature of oil supply shocks as reflected in the Cross-Asset Event Study chart below **(Img. 4)**. In equity-led stress WTI rallies persistently across all windows (+15.39% - +14.66% - +16.58%) with a 100% hit rate throughout, suggesting that equity-led convergence episodes are associated with genuine supply disruption fears or commodity inflation dynamics rather than demand destruction.

There is a 90-day mean reversion principle identified in the data which may still apply in reverse in this current supply disruption driven environment, just as historically oil crashed then recovered within 90 days during the demand destruction or supply glut driven oil-led episodes, the current elevated oil prices may revert lower as geopolitical risk premium unwinds. **The historical precedent suggests monitoring the 90-day window from the signal activation date as a key inflection point.**



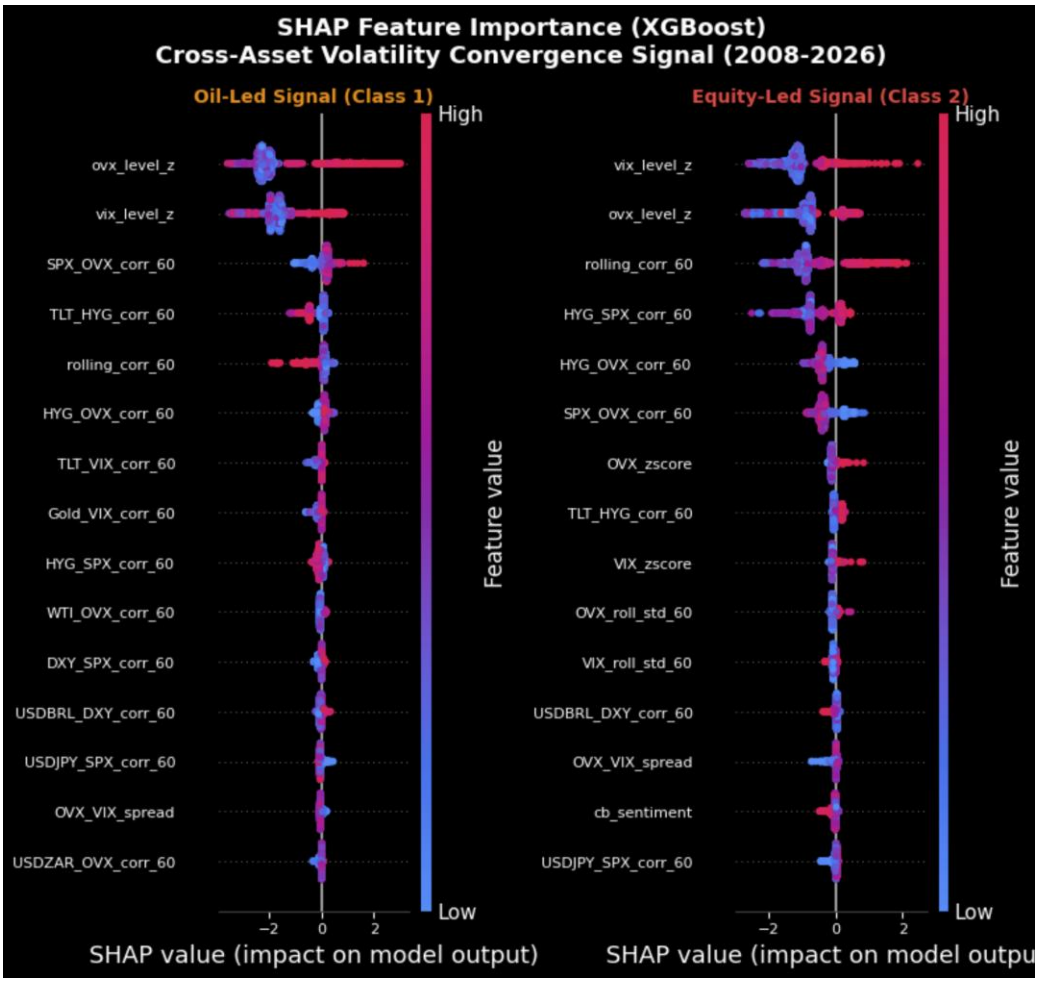
Img.4: Cross-Asset Event Study.

3.2 SHAP Feature Importance

XGBoost was trained on the full 2008 - 2026 dataset confirming the analytical framework through independent feature importance rankings:

- ovx_level_z and vix_level_z are the dominant predictors across both regimes, the Z-scores carry the primary predictive weight.
- rolling_corr_60 is 6x more important for equity-led signal activation, equity stress requires sustained co-movement (mean 0.697) while oil-led episodes feature OVX spiking independently (mean 0.393).
- HYG_SPX_corr_60 is 7x more important for equity-led stress, credit/equity coupling is the defining signature of financial market contagion.
- cb_sentiment shows asymmetric importance (0.036 equity-led vs 0.004 oil-led), equity stress is more policy-sensitive than oil supply shocks.
- OVX_VIX_spread has surprisingly low SHAP importance (0.039), the absolute level of both volatility measures matters more than their relative gap

Every methodological choice, signal construction, regime classification, CB sentiment integration, is confirmed by the model's feature importance rankings.



Img.5: SHAP Feature Importance.

3.3 XGBoost vs Logistic Regression

XGBoost significantly outperforms Logistic Regression, confirming the non-linear relationships between cross-asset features and regime-conditional signal activation. **Split 2 (2021 - 2022 test) produces the strongest results with an Oil-Led F1 of 0.97, Equity-Led F1 of 0.82 and overall average Oil-Led F1 of 0.81 vs Logistic Regression's 0.42**. The model is treated as an analytical validation tool rather than a production trading signal, with 228 convergence signal days across 18 years the dataset is too small for high-accuracy classification, the primary value is SHAP feature importance, not predictive accuracy..

4. Live Signal Assessment: Iran Crisis (2026)

The convergence signal activated on March 6th 2026 (VIX Z=1.09σ, OVX Z=3.80σ, correlation = 0.861), correctly identifying the Iran crisis oil-led stress environment. Two additional signal days on March 27 and March 30 confirmed the episode. The signal is currently inactive as of April 3 (VIX retreated to 0.45σ) but OVX remains elevated at 3.19σ.

Actual performance since March 6 signal activation:

Asset	Playbook Signal	Expected 30d	Actual (Mar 6 - Apr 3)	Verdict
WTI	SHORT*	-10.52%	+22.71%	Structural inversion
Brent	SHORT*	-8.73%	+17.63%	Structural inversion
Gold	LONG*	+2.13%	-9.61%	Liquidity flush
DXY	LONG*	+2.40%	+1.05%	Correct
USDCAD	LONG*	+3.35%	+1.81%	Correct
USDZAR	NEUTRAL	+2.34%	+1.88%	Correct
USDMXN	NEUTRAL	+3.63%	+0.82%	Correct
SPX	SHORT	-0.94%	-2.33%	Correct
NIKKEI	SHORT*	-4.67%	-5.68%	Correct
AUDUSD	SHORT*	-2.66%	-1.42%	Correct
USDCHF	SHORT*	-1.88%	+2.30%	SNB intervention
USDBRL	LONG	+3.11%	-2.03%	Oil exporter effect
TLT	LONG	+0.26%	-1.50%	Inflation fears

The adjusted directional accuracy of the Cross-Asset Playbook: 10/13 (77%) excluding WTI, Brent, USDBRL and USDCHF, all four with specific structural explanations documented in the analysis.

The supply disruption nature of the Iran war inverts the historical oil-led signal for WTI and Brent, consistent with the Russia-Ukraine parallel (Brent +42% at 30 days) within the dataset. Gold's -9.61% selloff mirrors the GFC 2008 pattern where initial liquidity flush (-11.08% at 30 days) preceded full recovery at 90 days. TLT's selloff reflects inflation fears from supply disruptions. If growth concerns dominate, the GFC playbook of TLT recovery may follow, if inflation continues to take centre stage and concerns heighten as the war continues at the 60-day mark, the Russia-Ukraine playbook could come into play.

The 90-day mean reversion principle, the most robust event study finding, points to a potential WTI retracement towards pre-crisis levels by June 2026 as geopolitical risk premium unwinds.

5. Limitations & Future Work

- Small signal sample: 228 convergence signal days across 18 years limits XGBoost and Logistic Regression models statistical power. Results should be interpreted as directional evidence rather than high-confidence predictions.
- FOMC sentiment only: ECB, BOJ and BOE sentiment scraping encountered JavaScript rendering limitations. Cross-CB sentiment coverage is identified as a key future enhancement.
- Supply disruption episodes: The historical oil-led playbook is conditioned predominantly on demand destruction and supply glut episodes. Supply disruption episodes (Russia-Ukraine, Iran war) require a separate conditioning framework.
- Signal sample size for LSTM: 14 convergence episodes is insufficient for reliable deep learning. LSTM was considered but excluded due to overfitting risk.
- Walk-forward test limitations: Liberation Day 2025 introduced a novel equity stress dynamic (tariff-driven) that the model had not encountered in training, resulting in collapsed equity-led recall in Split 3.
- All historical data is sourced via yfinance/Yahoo Finance. While sufficient for historical regime analysis, production implementation would use Bloomberg Data License or Refinitiv Eikon feeds for improved data quality, roll-adjusted futures series and standardised fixing times.

Future enhancements: multi-CB sentiment coverage, supply disruption regime sub-classification, options market data integration (25-delta risk reversals), Bloomberg/Refinitiv data source, real-time signal monitoring dashboard.

6. Conclusion

This project began with a Bloomberg chart, crude oil 25-delta call volatility and SPX put volatility converging as the Iran conflict took off in early 2026. The central question was simple: when oil and equity volatility converge above a critical threshold, does the nature of the leading stress determine the subsequent cross-asset response? The answer, validated across 18 years of data and 14 convergence episodes, is yes.

The convergence signal framework identifies two distinct stress regimes, oil-led and equity-led, each producing structurally different cross-asset responses. The regime distinction is not cosmetic, TLT strengthens in oil-led episodes and sells off in equity-led ones, Gold maintains a LONG* signal across both regimes, and the dollar strengthens persistently in oil-led stress. In demand destruction and supply glut driven oil-led episodes, WTI is a SHORT* (-10.52%), inverting to a LONG* with 100% hit rate in equity-led stress (+15.39%). Getting the regime wrong means executing the opposite trade. The playbook is regime-conditional by design.

The XGBoost model and SHAP analysis validate the analytical framework independently, ovx_level_z and vix_level_z are the dominant predictors, rolling correlation distinguishes equity-led from oil-led with 6x differential importance, and central bank sentiment confirms equity stress is more policy-sensitive than oil supply shocks.

The live Iran war test is where the framework delivers its most honest and most valuable output. The convergence signal activated on March 6th, 2026, correctly identifying an oil-led stress environment. Unlike the historical oil-led episodes which were predominantly demand destruction or supply glut driven, where WTI fell -10.52% at 30 days, the Iran war's supply disruption nature inverted the oil signal, with WTI rallying +22.71%. This mirrors the Russia-Ukraine parallel within the dataset where Brent rallied +42% at 30 days under identical supply disruption dynamics.

Adjusted directional accuracy of 77% across non-structurally distorted assets reflects the framework's effectiveness when structural deviations are properly accounted for. The 90-day mean reversion principle, the most robust event study finding, now points to a potential WTI retracement toward pre-crisis levels by June 2026 as geopolitical risk premium unwinds, at which point the historical playbook relationships may reassert themselves.

The framework does not predict markets. It identifies stress regimes, conditions historical playbook responses on those regimes, and flags where current episodes diverge from historical precedent and why. In the current Iran crisis, that capability identified the oil-led convergence, surfaced the Russia-Ukraine analogue, determined the trades to execute and recognised where to deviate from the standard playbook. That is precisely what the framework was built to do.

Full code, data and methodology available on GitHub (Israel Olujobi).

April 2026.